Denaturing SDS polyacrylamide gel electrophoresis

Risk assessment

– **Acrylamide is a KNOWN NEUROTOXIN and LIKELY CARCINOGENIC!**
– **Work with high voltage power sources**

▷ Wear gloves, safety glasses, lab coat

□ Collect acrylamide monomer solutions as HAZARDOUS WASTE

□ DO NOT wash into sewer

Reviewed: Jul 17, 2024

This is a bench card. Full protocol available online.

›› *Casting SDS polyacrylamide gels*

□ Gel casting apparatus
□ Gel cassette, 0.5–1.5 mm spacer
□ Gel comb, 12 wells
□ Isopropyl alcohol

(1.) Prepare resolving and stacking gel solutions in the order of the recipes below, but omit ammonium persulfate and TEMED. Invert the solution several times to mix the ingredients. Avoid foaming!

(2.) *Optional:* Filter the solutions through a 0.45 µm pore.

(3.) *Optional:* Degas under vacuum for 15 min at room temperature.

Quality assurance: For reproducible polymerization, the dissolved oxygen must be removed.

(4.) Assemble the gel cassette. Mark the front plate about 1 cm below the bottom of the comb to indicate the top of the resolving gel. Remove the comb.

(5.) Add ammonium persulfate and TEMED to the resolving gel solution, invert gently several times to mix. Work quickly since polymerization begins immediately.

(6.) Pour the resolving gel solution into the assembled gel cassette with the comb removed.

Gel fraction	Spacer	Mini Gel (7.2 cm × 8.6 cm)	Midi Gel (13.3 cm × 8.7 cm)
Resolving gel	1.0 mm	5.0 mL	9.0 mL
Stacking gel	1.0 mm	1.5 mL	2.5 mL

(7.) Gently overlay the resolving gel with a few drops of isopropyl alcohol to create a smooth interface.

(8.) Allow 45–90 min for the gel to polymerize. This ensures reproducible pore size. 60 min

Quality assurance: A sharp interface between the gel and the alcohol overlay indicates complete polymerization.

(9.) Pour off the alcohol, rinse with water, dry the area above the resolving gel by touching a strip of filter paper to the edge of the cassette.

(10.) Add ammonium persulfate and TEMED to the stacking gel solution. Invert gently to mix.

(11.) Pour the stacking gel solution in the gel cassette. To avoid spilling, place the pipet tip at an inclined angle and dispense slowly. Fill to the rim before inserting the comb.

(12.) Store gels flat at 4 °C with the comb inserted. Wrap in a wet paper towel and seal in a plastic bag.

Critical: Run handcast gels with discontinuous buffer systems right after casting since the discontinuity gradually disappears. Basic Tris-Cl SDS-PAGE gels must be used within a couple of weeks; Bis-Tris gels are stable for up to six months.

›› *Running SDS polyacrylamide gels*

☐ Electrophoresis tank	☐ 4 × sample buffer
☐ Power supply	☐ 20% Iodoacetamide, 1 mL

(1.) Remove comb by pulling straight up, slowly and gently. Rinse wells carefully with water.

(2.) Assemble the electrophoresis cell.

(3.) Add fresh running buffer to the reservoir that connects to the stacking gel. Fill the other reservoir with fresh or used running buffer.

> ***Critical:*** Always fill both inner and outer chambers with the recommended volume of running buffer. Incomplete filling may cause uneven band migration or excessive heating.

(4.) Prepare samples in 1 × sample buffer. Load 0.1–5 µg of purified protein or up to 25 µg crude extract per lane.

> ***Quality assurance:*** Ensure an excess of SDS; typically 1.5–3.0 µg SDS per microgram protein. Diluted sample buffer provides approximately 10 µg/µL SDS, which is sufficient for most applications up to 5 µg protein per lane.

(5.) Immediately heat to 80–90 °C for 2 min (5 min for pellets) to denature the proteins.

(6.) *Optional:* Bring to 60 °C, add iodoacetamide to 2% (w/v). Incubate for 30 min at room temperature.

(7.) Remove insoluble material at 17 000 × *g* for 2 min.

> ***Quality assurance:*** Cleared SDS-PAGE samples stored at 4 °C overnight or frozen for longer periods, should be briefly warmed at 37 °C to redissolve SDS. Remove insoluble material by centrifugation.

(8.) Load the appropriate volume of each sample per well. For midi gels, wells typically accommodate 15–40 µL. Use a long pipette tip for loading. Viscous samples often fail to settle into the well and can cause smears.

(9.) Close the lid, connect tank and power supply, perform electrophoresis at constant voltage. 30–60 min

- Run at 5–10 V per centimeter of gel (70–80 V for Mini and Midi gels) for about 10 min until the sample is concentrated at the starting point of the resolving gel.
- Increase the voltage for the resolving phase as recommended. Do not exceed the limits set by the manufacturer for the electrophoresis system.

(10.) Turn the power supply off and disconnect the electrical leads.

(11.) Pop open the gel cassette. Gently float the gel off the plate into a water basin for staining or transfer.

A › *Tris-Cl/Tris-glycine SDS PAGE (Laemmli)*

- ☐ R0102 0.5 M Tris-Cl, 100 mL
- ☐ R0103 1.5 M Tris-Cl, 250 mL
- ☐ R0104 4 × Laemmli sample buffer
- ☐ R0105 5 × Tris-Gly running buffer, 1.0 L

(1.) Prepare the gel solutions.

- Combine the ingredients below to prepare 24 mL resolving gel solution sufficient for two Midi gels (9 mL each) or four Mini gels (5 mL each) with 1.0 mm spacer. Select the column matching your desired acrylamide percentage.

  With 30% acrylamide-bis acrylamide solutions (other ingredients as below):

Ingredient	**Stock**	**Final**	**7.0%**	**8.0%**	**10.0%**	**12.0%**	**15.0%**	**16.0%**
Water, reagent-grade			12.2 mL	11.4 mL	9.8 mL	8.2 mL	5.8 mL	5.0 mL
Acrylamide/bis-acrylamide (37.5:1)	30%		5.6 mL	6.4 mL	8.0 mL	9.6 mL	12.0 mL	12.8 mL

  With 40% acrylamide-bis acrylamide solutions:

Ingredient	**Stock**	**Final**	**7.0%**	**8.0%**	**10.0%**	**12.0%**	**15.0%**	**16.0%**
Water, reagent-grade			13.6 mL	13.0 mL	11.8 mL	10.6 mL	8.8 mL	8.2 mL
Acrylamide/bis-acrylamide (37.5:1)	40%		4.2 mL	4.8 mL	6.0 mL	7.2 mL	9.0 mL	9.6 mL
Tris-Cl, pH 8.8	1.5 M	375 mM	6.0 mL	6.0 mL	6.0 mL	6.0 mL	6.0 mL	6.0 mL
SDS	20%	0.1%	120 µL	120 µL	120 µL	120 µL	120 µL	120 µL
TEMED	100%	0.1%	24 µL	24 µL	24 µL	24 µL	24 µL	24 µL
Ammonium persulfate (APS)	10%	0.1%	240 µL	240 µL	240 µL	240 µL	240 µL	240 µL

- For 8 mL stacking gel solution.

  With 30% acrylamide-bis acrylamide solutions (others below):

Ingredient	**Stock**	**Final**	**4.0%**	**5.0%**
Water, reagent-grade			4.8 mL	4.6 mL
Acrylamide/bis-acrylamide (37.5:1)	30%		1.1 mL	1.3 mL

  With 40% acrylamide-bis acrylamide solutions:

Ingredient	**Stock**	**Final**	**4.0%**	**5.0%**
Water, reagent-grade			5.1 mL	4.9 mL
Acrylamide/bis-acrylamide (37.5:1)	40%		0.8 mL	1.0 mL
Tris-Cl, pH 6.8	0.5 M	125 mM	2.0 mL	2.0 mL
SDS	20%	0.1%	40 µL	40 µL
TEMED	100%	0.1%	8 µL	8 µL
Ammonium persulfate (APS)	10%	0.1%	80 µL	80 µL

(2.) Dilute samples with 4 × Laemmli sample buffer.

(3.) Run gels in 1 × Tris-Gly running buffer at 180–200 V for 40–60 min.

B › *Bis-Tris/MES and Bis-Tris/MOPS SDS PAGE*

- ☐ R0106 3.5 × Bis-Tris, 1.0 L
- ☐ R0107 4 × LDS sample buffer, 10 mL
- ☐ R0108 20 × MES running buffer, 0.5 L
- ☐ R0109 20 × MOPS running buffer, 0.5 L
- ☐ 1 M Bisulfite, 5 mL

(1.) Prepare the gel solutions.

- Combine the ingredients below to prepare 24 mL resolving gel solution sufficient for two Midi gels (9 mL each) or four Mini gels (5 mL each) with 1.0 mm spacer. Select the column matching your desired acrylamide percentage.

  With 30% acrylamide-bis acrylamide solutions (other ingredients as below):

Ingredient	**Stock**	**Final**	**7.0%**	**8.0%**	**10.0%**	**12.0%**	**15.0%**	**16.0%**
Water, reagent-grade			11.4 mL	10.6 mL	9.0 mL	7.4 mL	5.0 mL	4.2 mL
Acrylamide/bis-acrylamide (37.5:1)	30%	*(var.)*	5.6 mL	6.4 mL	8.0 mL	9.6 mL	12.0 mL	12.8 mL

  With 40% acrylamide-bis acrylamide solutions:

Ingredient	**Stock**	**Final**	**7.0%**	**8.0%**	**10.0%**	**12.0%**	**15.0%**	**16.0%**
Water, reagent-grade			12.8 mL	12.2 mL	11.0 mL	9.8 mL	8.0 mL	7.4 mL
Acrylamide/bis-acrylamide (37.5:1)	40%	*(var.)*	4.2 mL	4.8 mL	6.0 mL	7.2 mL	9.0 mL	9.6 mL
Bis-Tris, pH 6.8	3.5 ×	360 mM	6.9 mL	6.9 mL	6.9 mL	6.9 mL	6.9 mL	6.9 mL
TEMED	100%	0.1%	24 µL	24 µL	24 µL	24 µL	24 µL	24 µL
Ammonium persulfate (APS)	10%	0.06%	144 µL	144 µL	144 µL	144 µL	144 µL	144 µL

- For 8 mL stacking gel solution.

  With 30% acrylamide-bis acrylamide solutions (others below):

Ingredient	**Stock**	**Final**	**4.0%**	**5.0%**
Water, reagent-grade			4.6 mL	4.4 mL
Acrylamide/bis-acrylamide (37.5:1)	30%	*(var.)*	1.1 mL	1.3 mL

  With 40% acrylamide-bis acrylamide solutions:

Ingredient	**Stock**	**Final**	**4.0%**	**5.0%**
Water, reagent-grade			4.9 mL	4.7 mL
Acrylamide/bis-acrylamide (37.5:1)	40%	*(var.)*	0.8 mL	1.0 mL
Bis-Tris, pH 6.8	3.5 ×	360 mM	2.3 mL	2.3 mL
TEMED	100%	0.1%	8 µL	8 µL
Ammonium persulfate (APS)	10%	0.06%	48 µL	48 µL

(2.) Dilute samples with 4 × Laemmli sample buffer or 4 × LDS sample buffer.

(3.) Typical run conditions: 30–60 min

- To resolve 5–75 kDa proteins, 1 × MES running buffer, 200 V for 35 min, 4 °C.
- To resolve 50–200 kDa proteins, 1 × MOPS running buffer, 200 V for 50 min, room temperature.

> ***Critical:*** Replace running buffer as soon as the pH indicator in the LDS sample buffer turns yellow.

> ***Critical:*** Reducing agents such as 2-mercaptoethanol (BME) and dithiothreitol (DTT) are not ionized in Bis-Tris and therefore do not enter the gel. To prevent protein reoxidation during electrophoresis, add 5 mM bisulfite (final) to the running buffer.

Recipe (available online) Resources (available online) Troubleshooting (available online) Notes (available online)